MECHANICAL DESIGN ENGINEER · LAZIO, ITALY

MAHDI BACHRAOUI

Drawings ready for the shop floor, with ISO GPS tolerancing and design reviews behind them.

Selected work from a nine-sheet portfolio. Full set with every released drawing: mahdibachraoui.github.io

CSWP

Certified SolidWorks Professional

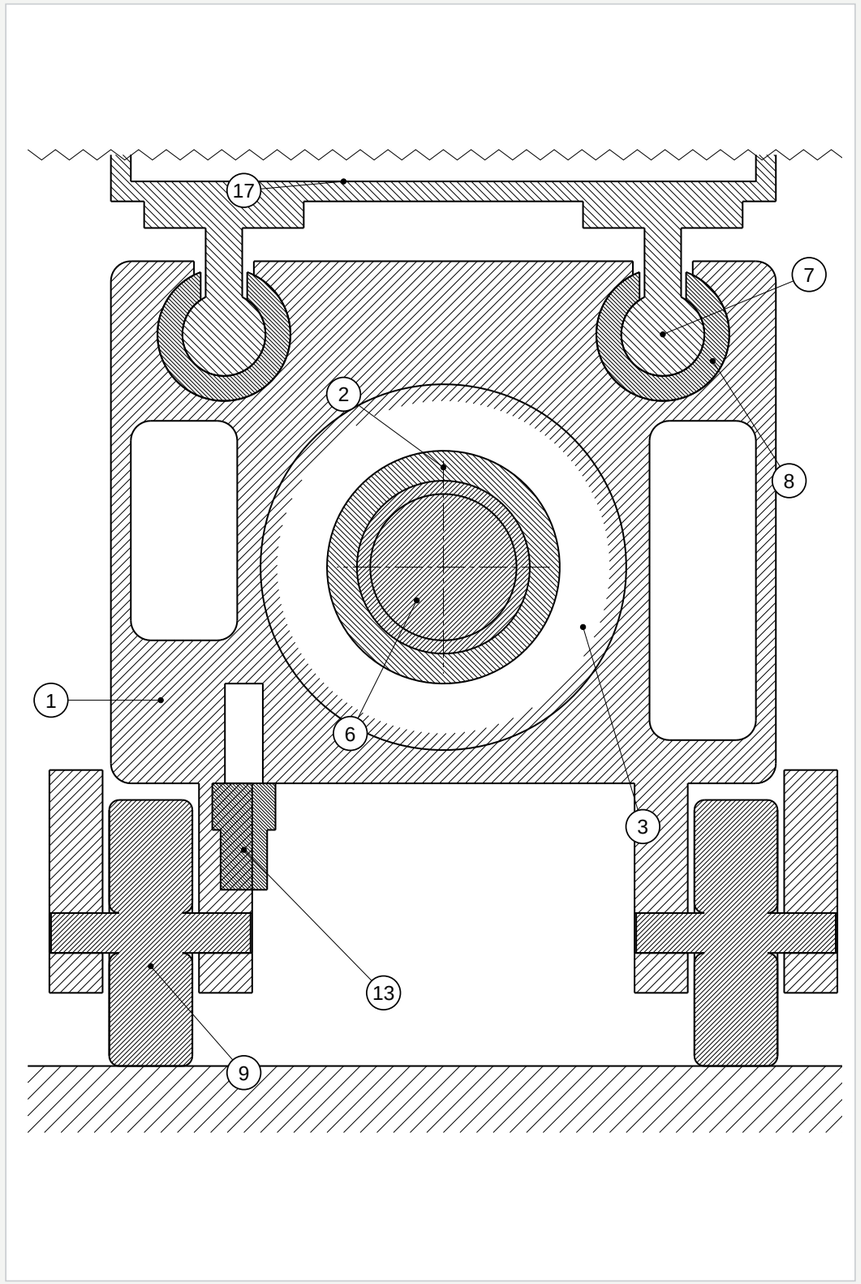
ISO GPS

ISO 1101, 5459, 2692, 8015

AUDI

Harness change management, Kromberg & Schubert

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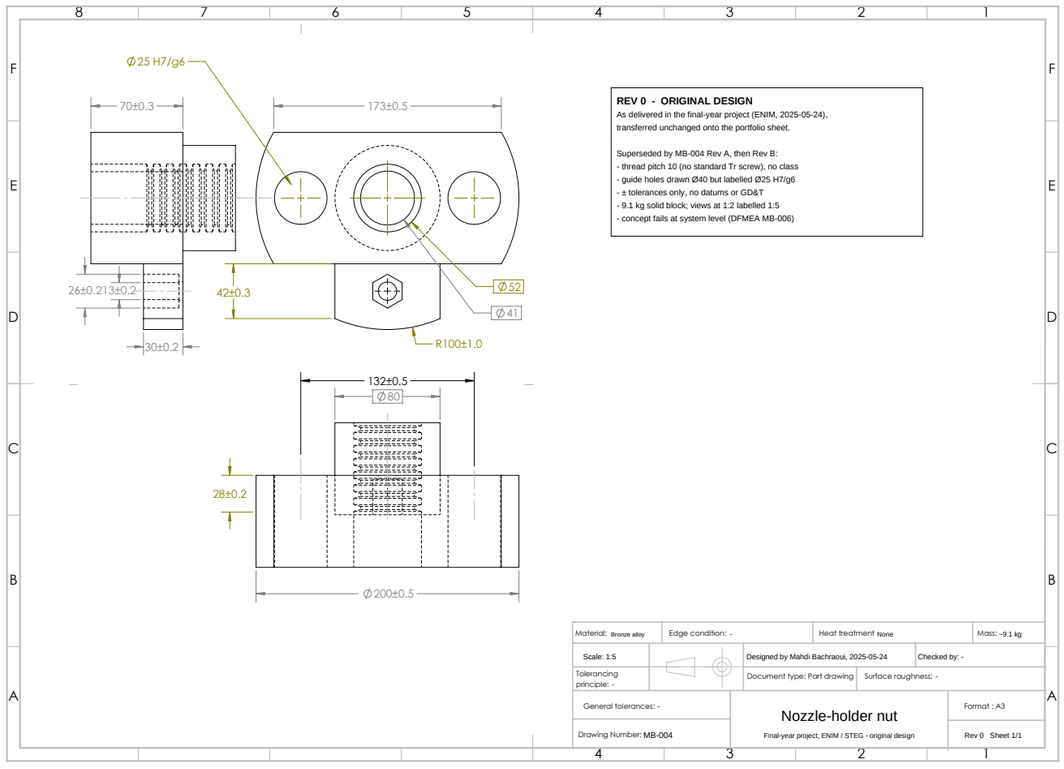
MB-005 · SECTION A-A · 1:2

MB-004 · REV B

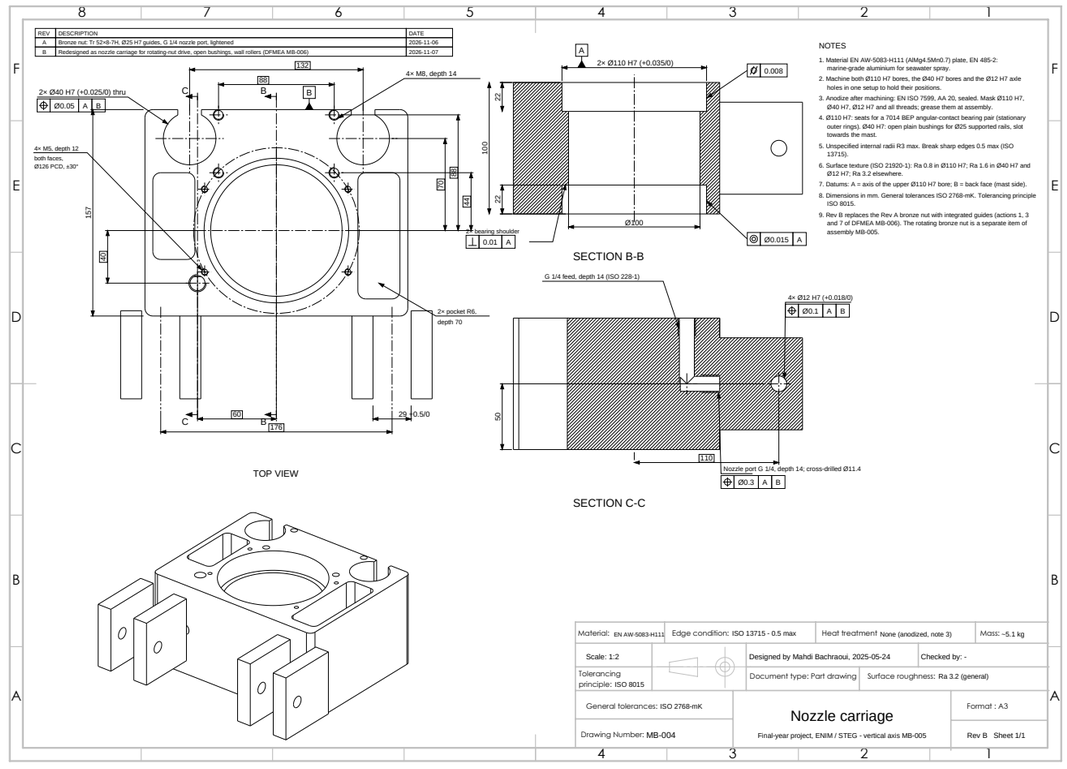
NOZZLE CARRIAGE

Item 1 of assembly MB-005, EN AW-5083-H111. It started as a 9.1 kg bronze nozzle-holder nut.

REV 0 · ORIGINAL



AFTER · REV B



PROBLEM

Rev 0 used a pitch-10 thread with no standard screw, guide holes drawn Ø40 but labelled Ø25, and only ± tolerances. Rev A fixed the part (Tr 52×8-7H, Ø25 H7 guides, G 1/4 port, lightening pockets). The DFMEA then showed the concept itself failed, so Rev B turns the nut into a carriage.

WHAT CHANGED

- Marine-grade EN AW-5083-H111: 5.14 kg instead of 9.1 kg.
- Two Ø110 H7 seats for the 7014 bearing pair that carries the rotating nut.
- Open Ø40 H7 bushing bores for supported rails; four Ø12 H7 axles for the wall rollers.
- All precision bores machined in one setup; anodised AA 20 with the fits masked.

STANDARDS AND SKILLS

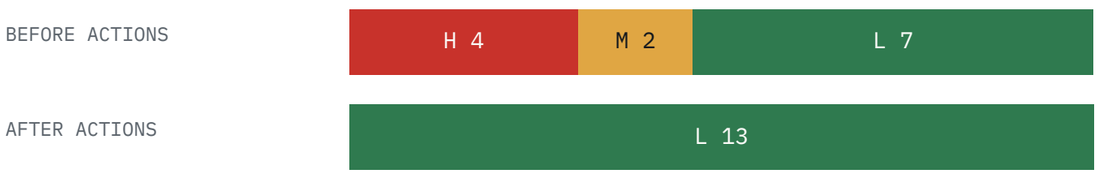
ISO 1101 · ISO 2901 · ISO 228-1 · EN ISO 7599 · position tolerancing · machining

MB-006 · REV A

VERTICAL AXIS DFMEA

AIAG-VDA (2019) design FMEA with Action Priority on the original vertical-axis concept. 13 failure chains, 2 sheets, with an Excel version.

ACTION PRIORITY · 13



PROBLEM

The original concept had never been checked at system level. Four failure chains came out with high Action Priority.

WHAT CHANGED

- Each action points to a feature in MB-004 Rev B or MB-005.
- Seven actions are designed and marked "test pending": only a test closes them.
- Ratings after actions are design targets, not test results.

STANDARDS AND SKILLS

AIAG-VDA 2019 · severity, occurrence, detection · Action Priority · Excel + PDF

HIGH PRIORITY · 4

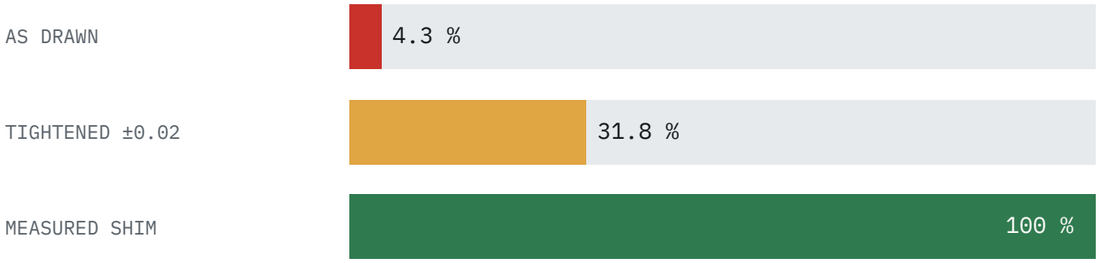
FAILURE MODE	S/O/D BEFORE	ACTION	S/O/D AFTER
Screw whirls above its critical speed	8 / 9 / 8	Rotating-nut drive; screw fixed and pre-tensioned	8 / 2 / 3
Guide rods deflect under the jet	7 / 10 / 7	Rollers carry the jet reaction to the wall	7 / 2 / 3
Nozzle blows out of the G 1/4 port	9 / 4 / 3	Parallel-thread nozzle with bonded seal; 375 bar proof test	9 / 2 / 2
High-pressure hose bursts	9 / 4 / 4	Energy chain, whip restraints at both ends	9 / 2 / 3

MB-007 + MB-005-02 · REV A

TOLERANCE STACK-UP AND ROTATING NUT

Can the drawn tolerances hold the bearing preload? Worst case, RSS and Monte Carlo on the real drawing values. Two sheets.

BEARING PRELOADS IN WINDOW · MONTE CARLO



PROBLEM

General tolerances were enough to draw the carriage, but nothing proved they could set the preload on the 7014 bearing pair. The window is only 0.03 mm.

WHAT CHANGED

- Tightening the machined parts does not solve it: bearing width tolerance dominates.
- Adopted: measure at assembly and grind one shim. MB-004 stays as drawn.
- Rotating nut drawn with a 97.5 mm clamp length, so the shim is always positive.
- Design study: the preload window and bushing bore are assumptions marked on the sheet.

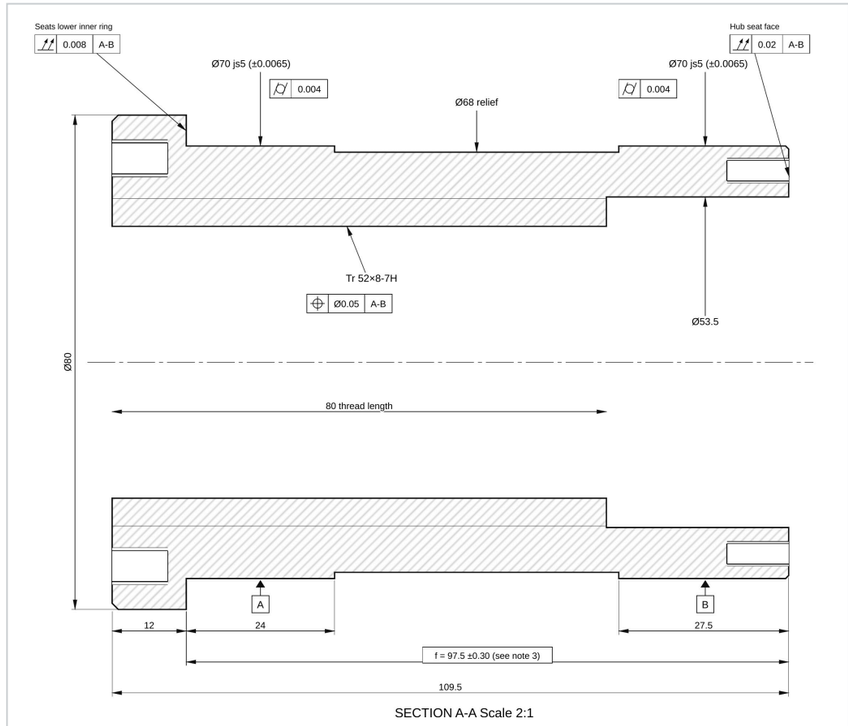
STANDARDS AND SKILLS

worst case · RSS · Monte Carlo · ISO 492 · ISO 2768 · ISO 2901 · js5 · total run-out

PRELOAD δ · MM

CASE	MEAN	WORST CASE
As drawn	−0.150	−1.300 to +1.000
Tightened	+0.020	−0.170 to +0.210
Measured shim	+0.020	+0.005 to +0.035

MB-005-02 ROTATING NUT · SECTION A-A



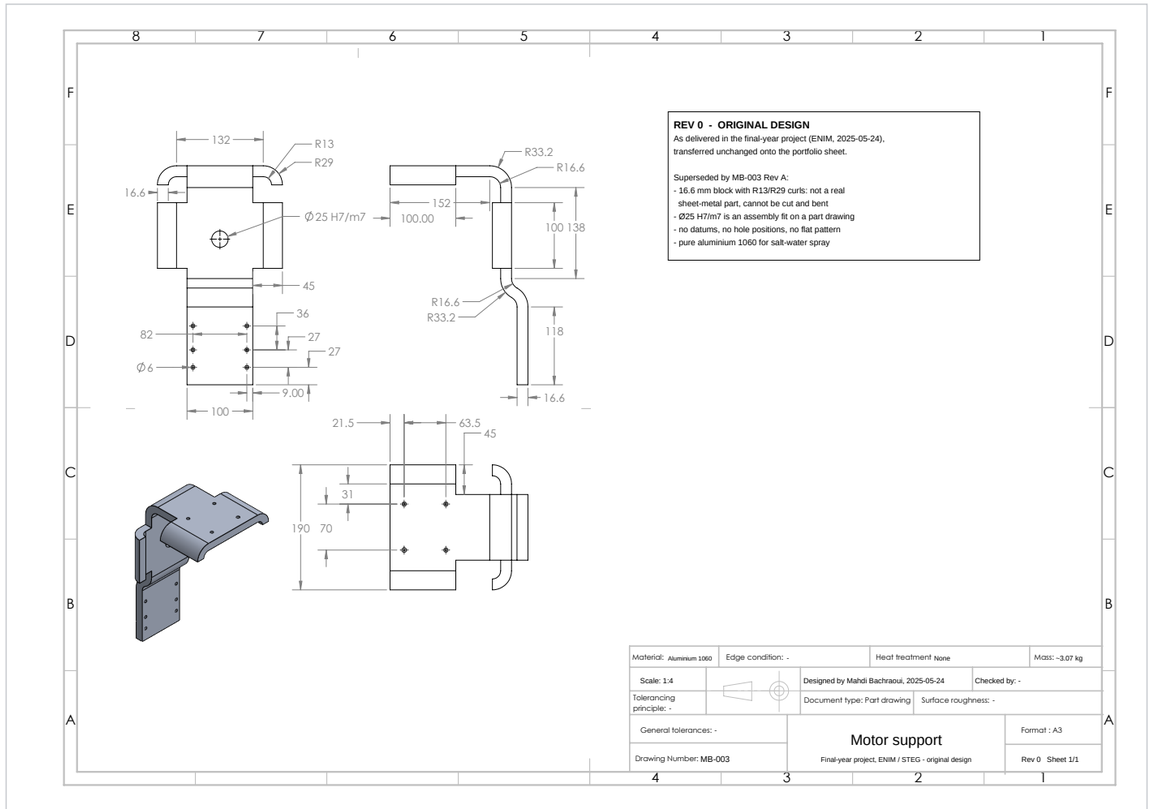
SECTION A-A Scale 2:1

MB-003 · REV A

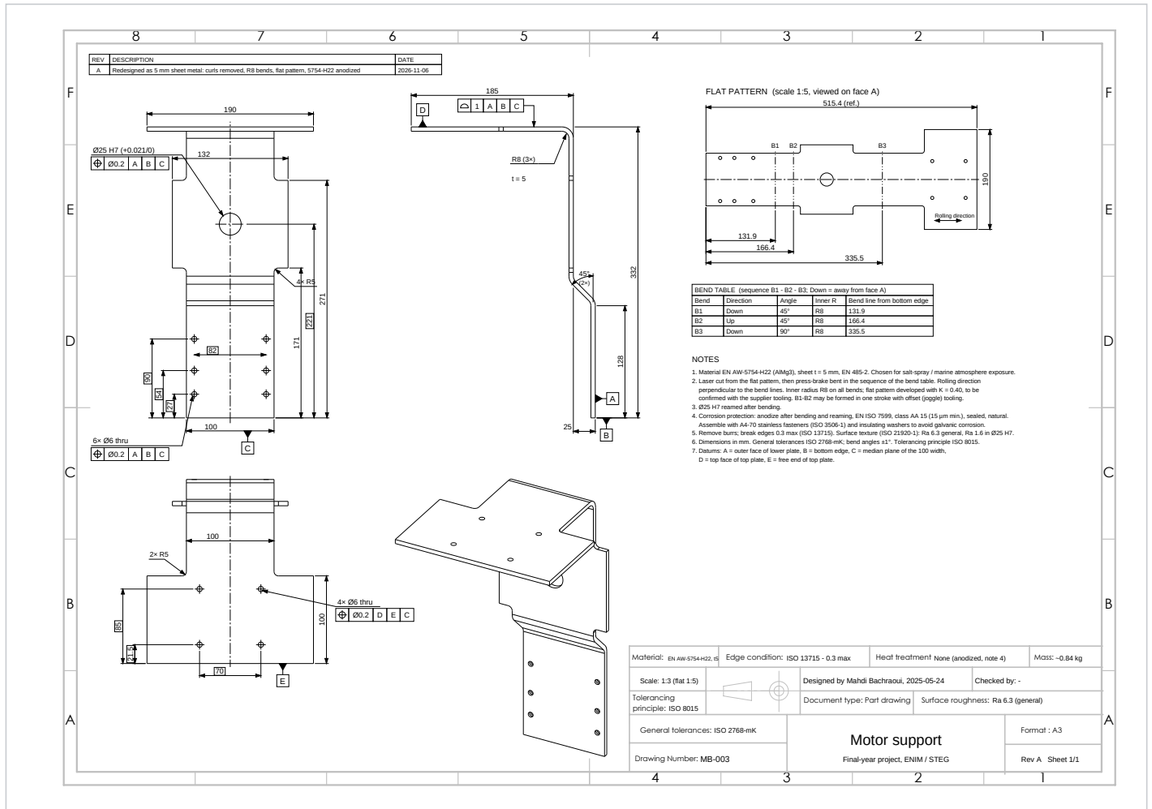
MOTOR SUPPORT

Bracket carrying two motors in salt-water spray. Final-year project, STEG Radès.

REV 0 · ORIGINAL



AFTER · REV A



PROBLEM

The original was a 16.6 mm block with rolled curls in pure aluminium 1060. It could not be laser cut and press-brake bent, and 1060 is a weak choice near seawater.

WHAT CHANGED

- Redesigned as 5 mm EN AW-5754-H22 sheet: laser cut, three press-brake bends at R8.
- Flat pattern (K = 0.40) and bend table with sequence; rolling direction across the bends.
- Ø25 H7 reamed after bending; datums A to E for both motor plates.
- Anodised AA 15 sealed; A4-70 fasteners with insulating washers against galvanic corrosion.

STANDARDS AND SKILLS

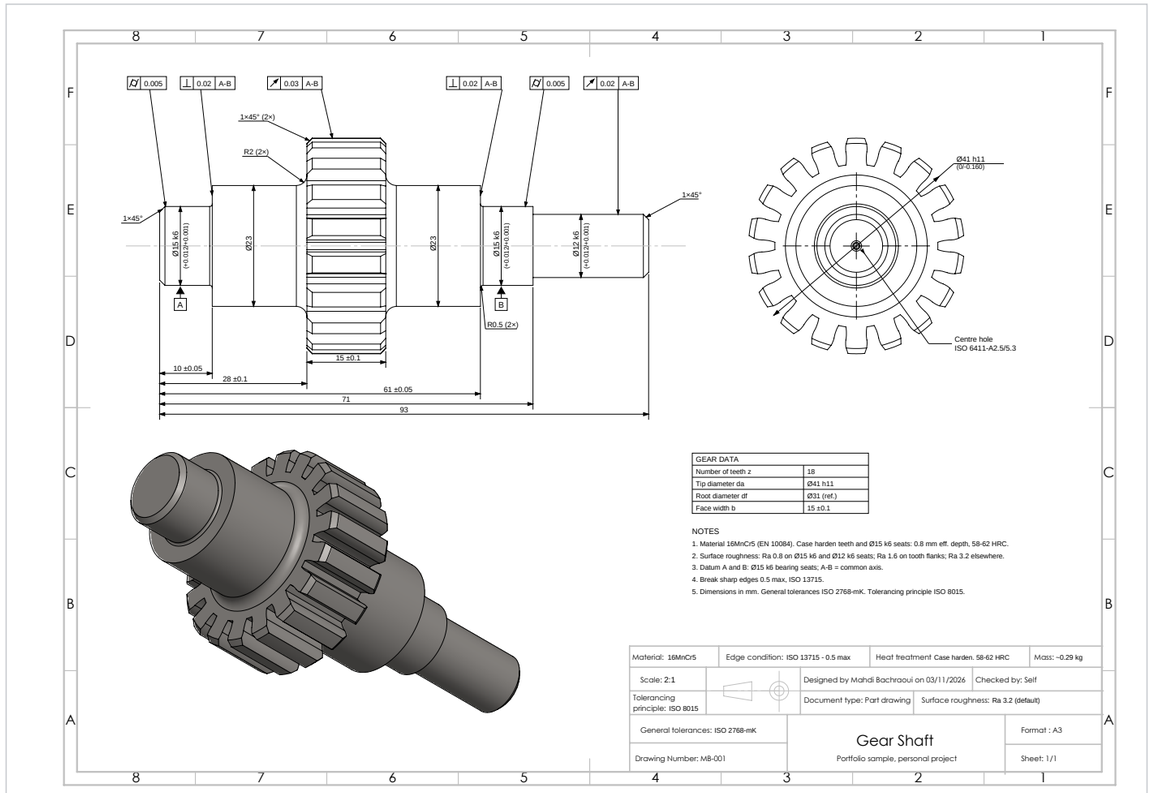
EN 485-2 · EN ISO 7599 · ISO 3506-1 · ISO 13715 · sheet-metal design · corrosion

MB-001 · REV A

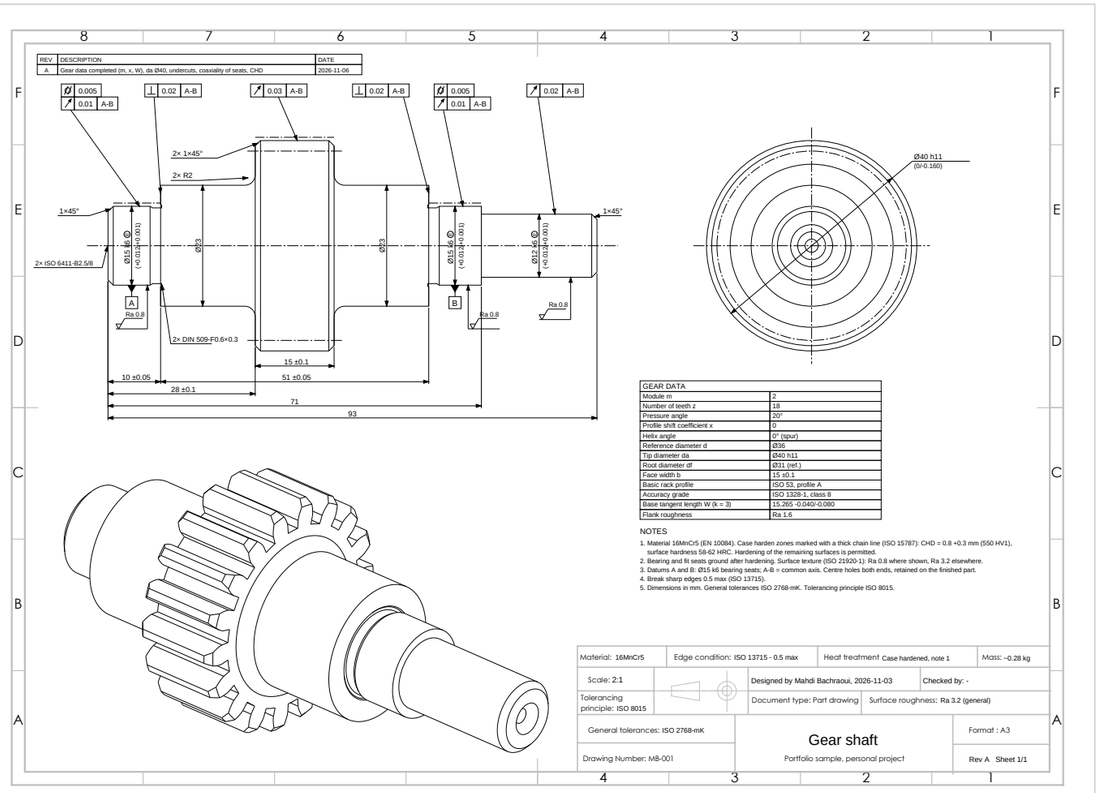
GEAR SHAFT

Case-hardened spur pinion shaft in 16MnCr5, m 2, z 18, between two Ø15 k6 bearing seats. Personal project.

BEFORE



AFTER · REV A



PROBLEM

The original sheet gave only four gear values, a tip diameter that did not match d + 2m, no undercuts at the ground seats and no way to show where the part is case hardened.

WHAT CHANGED

- Full gear data block: module, pressure angle, profile shift, ISO 1328-1 class 8 and base tangent length W (k = 3) for inspection.
- Tip diameter corrected from Ø41 to Ø40 h11 (d + 2m).
- DIN 509 undercuts so the k6 seats can be ground; envelope requirement (E) and run-out to the common axis A-B.
- Case-hardened zones marked to ISO 15787: CHD 0.8 +0.3 mm at 550 HV1. Centre holes kept at both ends.

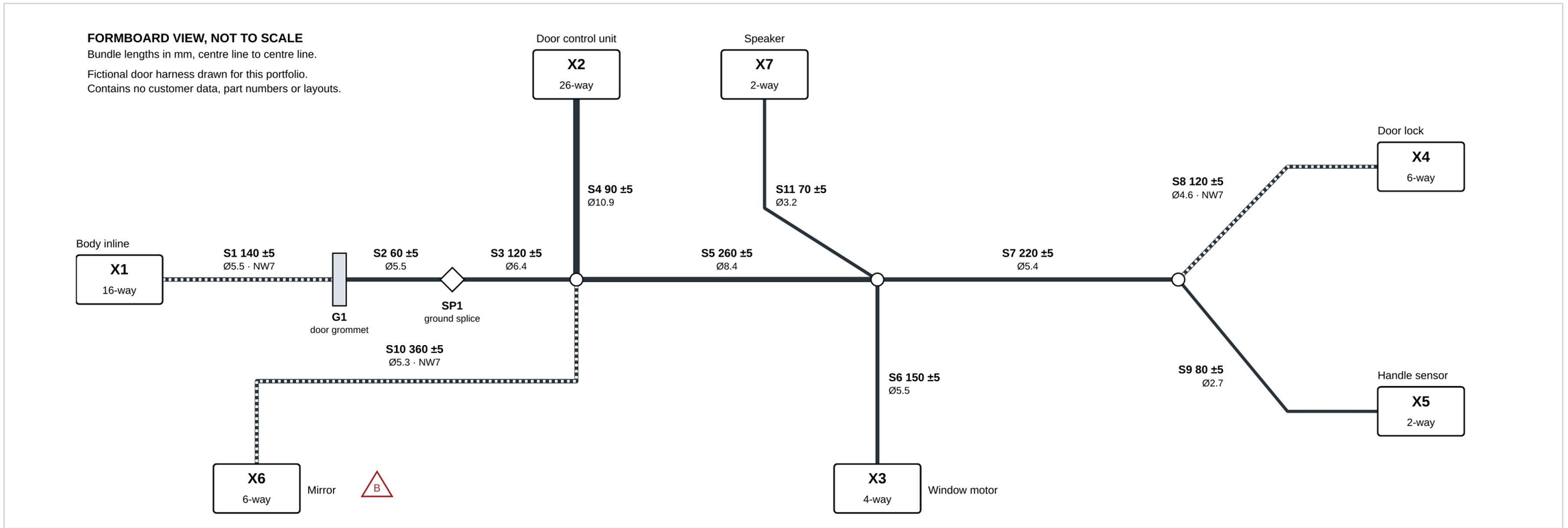
STANDARDS AND SKILLS

ISO 1101 · ISO 2203 · ISO 1328-1 · ISO 15787 · ISO 21920-1 · DIN 509 · gear geometry

MB-008 · REV B

DOOR HARNESS

A fictional automotive door harness in the working format of harness engineering. It contains no customer data.



WHAT IT SHOWS

- Every wire length summed from its route over the bundle segments.
- Bundle diameters and conduit sizes calculated from the wires inside.
- Change CR-0042 implemented as Rev B: new circuits flagged, affected bundles re-sized.

STANDARDS AND SKILLS

ISO 6722-1 · FLRY-B · wire list · change management

22
circuits

11
bundle segments

7
connectors

0.21 V
window motor drop at an assumed 15 A

WORK WITH ME

Open to mechanical design roles in Italy and the EU, and to freelance drawing, GD&T and design-review work.

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